Wafeeq Platform — Backend API Phase Plan (4 Months)

Scope: Backend API only (SOW: Backend API & UI/UX Design).
Duration: 4 months (16 weeks).
Reference: SOW Document, February 04, 2026.

Phase template

Every phase uses the same structure:

Item	Content
Phase	Number and name
Description	High-level scope of the phase
Timeline	Week range
Technical	Tech stack / implementation notes for the phase
DB models	Table/model names only
Sub-phases	Numbered breakdown (e.g. 1.1, 1.2)

Overview

Phase	Focus	Timeline
Phase 1	Foundation & authentication	Weeks 1–4 (Month 1)
Phase 2	Core: courses, enrollments, trainer applications	Weeks 5–6 (Month 2)
Phase 3	**Dictionary** (separate phase)	Weeks 7–8 (Month 2)
Phase 4	**Academic grade system** (separate phase)	Weeks 9–10 (Month 3)
Phase 5	E-commerce, admin, Node scaling	Weeks 11–12 (Month 3)
Phase 6	Testing & documentation	Weeks 13–14 (Month 4)
Phase 7	Deployment (DigitalOcean, K8s)	Weeks 15–16 (Month 4)

Phase 1 — Foundation & Authentication

Description
Set up the API foundation: project structure, database, migrations, and authentication.

Delivers JWT-based auth (access and refresh tokens), RBAC, and user management. Establishes `/api/v1/`, standardized error handling, validation, rate limiting, and logging.

Timeline
Weeks 1–4 (Month 1).

Technical
Node.js, Express; JWT (access + refresh), bcrypt/Argon2; PostgreSQL (or chosen DB) + migrations; express-validator/Joi; helmet, cors, rate-limit; Winston/Pino logging; env-based config.

DB models
Users, UserProfiles, RefreshTokens, PasswordResets.

Sub-phases

- **1.1** — Project setup: API versioning (`/api/v1/`), environment config, logging, security middleware (CORS, rate limiting, security headers).
- **1.2** — Database setup: PostgreSQL (or chosen DB), migrations, base schema for User and auth-related tables.
- **1.3** — Auth API: register, login, logout, refresh-token, verify-email, forgot-password, reset-password.
- **1.4** — user profile and preferences endpoints; RBAC (Student, Trainer, Admin).
- **1.5** — **Testing:** Unit tests for auth services; integration tests for register, login, refresh, forgot/reset password; RBAC checks.

endpoints

Method	Endpoint	Description
POST	`/api/v1/auth/register`	User registration
POST	`/api/v1/auth/login`	User login
POST	`/api/v1/auth/logout`	User logout
POST	`/api/v1/auth/refresh-token`	Refresh access token
POST	`/api/v1/auth/verify-email`	Email verification
POST	`/api/v1/auth/forgot-password`	Request password reset
POST	`/api/v1/auth/reset-password`	Reset password

User Profile

Method	Endpoint	Description
GET	`/api/v1/users/profile`	Get user profile
PUT	`/api/v1/users/profile`	Update profile
PUT	`/api/v1/users/change-password`	Change password
DELETE	`/api/v1/users/account`	Delete account
GET	`/api/v1/users/{id}`	Get user by ID

Method	Endpoint	Description
GET	`/api/v1/users/preferences`	Get preferences
PUT	`/api/v1/users/preferences`	Update preferences

Phase 2 — Core: Courses, Enrollments, Trainer Applications

Description
Implement course management (CRUD, sections, lessons), enrollment and progress tracking, and the trainer application workflow (apply, review, approve/reject). Dictionary is in Phase 3; academic grades are in Phase 4.

Timeline
Weeks 5–6 (Month 2).

Technical
Express routes/services/repositories; Multer + DigitalOcean Spaces (or S3) for course/lesson media and trainer application uploads; role middleware (educator, admin); pagination and filters (query params).

DB models
Courses, CourseSections, CourseLessons, CourseCategories, CourseEnrollments, CourseProgress, CourseReviews; TrainerApplications, TrainerProfiles (or equivalent).

Sub-phases

- **2.1** — Courses API: list, get, create, update, delete, submit-for-approval; sections and lessons; categories; featured/popular.
- **2.2** — Enrollments API: enroll, my-courses, progress, complete-lesson, certificate, unenroll.
- **2.3** — Trainer application API: apply, status, file uploads; admin APIs: list applications, approve, reject, start-review, schedule interview.
- **2.4** — **Testing:** Unit tests for course, enrollment, and application services; integration tests for CRUD, enroll, complete-lesson, application workflow.

API Endpoints

Course Management

Method	Endpoint	Description
GET	`/api/v1/courses`	List all courses (with filters)
GET	`/api/v1/courses/{id}`	Get course details
POST	`/api/v1/courses`	Create course (Trainer)
PUT	`/api/v1/courses/{id}`	Update course
DELETE	`/api/v1/courses/{id}`	Delete course
POST	`/api/v1/courses/{id}/submit-for-approval`	Submit for approval

Method	Endpoint	Description
GET	`/api/v1/courses/featured`	Get featured courses
GET	`/api/v1/courses/popular`	Get popular courses
GET	`/api/v1/courses/categories`	Get all categories

Course Sections & Lessons

Method	Endpoint	Description
GET	`/api/v1/courses/{id}/sections`	Get course sections
POST	`/api/v1/courses/{id}/sections`	Create section
PUT	`/api/v1/courses/{id}/sections/{sectionId}`	Update section
DELETE	`/api/v1/courses/{id}/sections/{sectionId}`	Delete section
POST	`/api/v1/courses/{id}/lessons`	Create lesson
PUT	`/api/v1/courses/{id}/lessons/{lessonId}`	Update lesson
DELETE	`/api/v1/courses/{id}/lessons/{lessonId}`	Delete lesson

Course Reviews

Method	Endpoint	Description
GET	`/api/v1/courses/{id}/reviews`	Get course reviews
POST	`/api/v1/courses/{id}/reviews`	Add review

Enrollment & Progress

Method	Endpoint	Description
POST	`/api/v1/enrollments`	Enroll in course
GET	`/api/v1/enrollments/my-courses`	Get my courses
GET	`/api/v1/enrollments/{courseId}/progress`	Get progress
PUT	`/api/v1/enrollments/{courseId}/progress`	Update progress
POST	`/api/v1/enrollments/{courseId}/complete-lesson`	Complete lesson
GET	`/api/v1/enrollments/{courseId}/certificate`	Get certificate
POST	`/api/v1/enrollments/{co`	Unenroll from course

Method	Endpoint	Description
	`urseId}/unenroll`	
GET	`/api/v1/enrollments/statistics`	Get enrollment stats

Trainer Applications

Method	Endpoint	Description
POST	`/api/v1/trainers/apply`	Submit trainer application
GET	`/api/v1/trainers/{id}`	Get trainer profile
GET	`/api/v1/trainers/search`	Search trainers
GET	`/api/v1/trainers/featured`	Get featured trainers

Admin - Trainer Applications

Method	Endpoint	Description
GET	`/api/v1/admin/trainer-applications`	List all applications
GET	`/api/v1/admin/trainer-applications/pending`	Pending applications
GET	`/api/v1/admin/trainer-applications/{id}`	Get application details
PUT	`/api/v1/admin/trainer-applications/{id}/approve`	Approve application
PUT	`/api/v1/admin/trainer-applications/{id}/reject`	Reject application
POST	`/api/v1/admin/trainer-applications/bulk-action`	Bulk approve/reject

Phase 3 — Dictionary (Separate Phase)

Description
Sign language dictionary: search and browse, categories, countries and world map, Word of the Day, favorites and search history, sign details (meanings, related signs, videos). Dictionary admin is in Phase 5 (Admin).

Timeline
Weeks 7–8 (Month 2).

Technical
Full-text or keyword search, pagination, filters (category, country); video URLs (multiple angles) via storage (Spaces/S3); Redis optional for Word of the Day and popular signs cache.

DB models
Signs, SignMeanings, SignVideos, SignCategories, SignCountries, Countries, CountryRegions, UserFavorites, SearchHistory, WordOfTheDay, RelatedSigns, SignLanguageGIFs.

Sub-phases

- **3.1** — Dictionary search: query param search, filters (category, country), pagination, sort (recent, popular).
- **3.2** — Dictionary browse: list signs (paginated), categories list and tree, alphabetical index by letter.
- **3.3** — Countries and world map: list countries with sign language, country profile by code, country-specific signs, regional variations, compare signs (2–4 countries).
- **3.4** — Word of the Day: get current, archive (by date range or calendar), track views/shares.
- **3.5** — User interactions: add/remove favorites, list user favorites; track and list search history per user.
- **3.6** — Sign detail: single sign by id; meanings (all), usage examples, related signs; videos (multiple angles: front, side, top).
- **3.7** — Sign language GIF tooltips: list all entries (e.g. 65), get by key, list by category.
- **3.8** — **Testing:** Unit tests for dictionary services; integration tests for search, browse, Word of the Day, favorites, sign detail.

API Endpoints

Dictionary Search & Browse

Method	Endpoint	Description
GET	`/api/v1/dictionary/search`	Search signs (q, category, country params)
GET	`/api/v1/dictionary/signs`	List all signs (paginated)
GET	`/api/v1/dictionary/signs/{id}`	Get sign details
GET	`/api/v1/dictionary/categories`	Get all categories
GET	`/api/v1/dictionary/signs/recent`	Recently added signs
GET	`/api/v1/dictionary/signs/popular`	Popular signs
GET	`/api/v1/dictionary/alphabetical/{letter}`	Browse by letter

World Map & Countries

Method	Endpoint	Description
GET	`/api/v1/dictionary/countries`	All countries with sign language

Method	Endpoint	Description
GET	`/api/v1/dictionary/countries/{code}`	Country profile
GET	`/api/v1/dictionary/countries/{code}/signs`	Country-specific signs
POST	`/api/v1/dictionary/signs/{id}/compare`	Compare signs (2-4 countries)
GET	`/api/v1/dictionary/regions/{country_code}`	Regional variations

Word of the Day

Method	Endpoint	Description
GET	`/api/v1/dictionary/word-of-the-day`	Get current word of day
GET	`/api/v1/dictionary/word-of-the-day/archive`	Archive (by date range)
POST	`/api/v1/dictionary/word-of-the-day`	Schedule word (Admin)

User Interactions

Method	Endpoint	Description
POST	`/api/v1/dictionary/favorites/{signId}`	Add to favorites
DELETE	`/api/v1/dictionary/favorites/{signId}`	Remove from favorites
GET	`/api/v1/dictionary/favorites`	List user favorites
POST	`/api/v1/dictionary/history/{signId}`	Track search
GET	`/api/v1/dictionary/history`	Get search history

Sign Details

Method	Endpoint	Description
GET	`/api/v1/dictionary/signs/{id}/meanings`	All meanings
GET	`/api/v1/dictionary/signs/{id}/usage-examples`	Usage examples
GET	`/api/v1/dictionary/signs/{id}/related-signs`	Related signs
GET	`/api/v1/dictionary/signs/{id}/videos`	Videos (multiple angles)

Sign Language GIF Tooltips

Method	Endpoint	Description
GET	`/api/v1/tooltips`	All 65 entries
GET	`/api/v1/tooltips/{key}`	Get by key
PUT	`/api/v1/tooltips/{key}`	Update (Admin)
POST	`/api/v1/tooltips/{key}/upload-gif`	Upload GIF (Admin)
DELETE	`/api/v1/tooltips/{key}/gif`	Delete GIF (Admin)
GET	`/api/v1/tooltips/categories`	List by category

Phase 4 — Academic Grade System (Separate Phase)

Description
Grade-level learning system (Module 8): 15 levels from Kindergarten (KG1–KG2) through Grades 1–12 and Adult. Delivers grade-wise curriculum, modules, lessons, enrollment and progress, grade assessments (submit, results, placement test), completion and certificates, promotion with prerequisites, and parent/guardian access for KG–Grade 8. Admin APIs for grade curriculum, lessons, and analytics are part of Phase 5.

Curriculum levels (structure only; content is client-provided)

- **Kindergarten (KG1–KG2), age 4–6:** Basic signs (colors, shapes, animals, family), greetings, emotions, numbers 1–10, basic alphabet.
- **Primary (Grades 1–5), age 6–11:** Grade 1 (daily routines, classroom, basic verbs); Grade 2 (food, clothing, body, simple sentences); Grade 3 (school subjects, hobbies, weather, time); Grade 4 (transportation, community, needs); Grade 5 (sports, nature, emotions, storytelling).
- **Middle (Grades 6–8), age 11–14:** Grade 6 (social, technology, health); Grade 7 (careers, science, geography); Grade 8 (advanced conversation, debate, abstract concepts).
- **High school (Grades 9–12), age 14–18:** Grade 9 (academic subject signs); Grade 10 (professional, business); Grade 11 (literature, culture); Grade 12 (academic discourse, interview, workplace).
- **Adult/Professional:** Workplace, medical, legal, technical/specialized terminology.

Timeline
Weeks 9–10 (Month 3).

Technical
Grade level enum or config (KG1, KG2, Grade1–12, Adult); prerequisite graph (e.g. Grade2 requires Grade1); placement test logic (score → suggested grade); certificate generation (PDF or link); parent–student linking with permission checks (KG–Grade 8).

DB models
GradeLevels, GradeCurriculum, GradeModules, GradeLessons, StudentGradeEnrollments,

GradeProgress, GradeAssessments, GradeAssessmentQuestions, GradeAssessmentResults, GradeCertificates, GradePrerequisites, ParentGuardianLinks.

Sub-phases

- **4.1** — Grade structure API: list grade levels (KG1, KG2, Grade1–12, Adult); get grade by level; curriculum, modules, lessons per grade.
- **4.2** — Enrollment and progress: enroll in grade, my-current-grade, my-progress, update completion, completion-status; prerequisites check.
- **4.3** — Grade assessments: list assessments per grade; submit assessment; get results; placement test (determine starting grade).
- **4.4** — Completion and advancement: complete grade, get certificate, promotion to next grade (with prerequisite validation).
- **4.5** — Parent/guardian access (KG–Grade 8): view student progress, report card; guardian-link (create/delete). Admin grade management (curriculum, lessons CRUD, analytics, completion rates, students-by-grade) is implemented in Phase 5.
- **4.6** — **Testing:** Unit tests for grade services; integration tests for enrollment, progress, assessments, placement test, completion, certificate.

API Endpoints

Grade Structure

Method	Endpoint	Description
GET	`/api/v1/grades`	List all grade levels (KG1-12, Adult)
GET	`/api/v1/grades/{level}`	Get grade by level
GET	`/api/v1/grades/{level}/curriculum`	Get complete curriculum
GET	`/api/v1/grades/{level}/modules`	Get learning modules
GET	`/api/v1/grades/{level}/lessons`	Get lessons

Enrollment & Progress

Method	Endpoint	Description
POST	`/api/v1/grades/{level}/enroll`	Enroll in grade
GET	`/api/v1/grades/my-current-grade`	Get current grade
GET	`/api/v1/grades/my-progress`	Get my progress
PUT	`/api/v1/grades/{level}/progress`	Update completion
GET	`/api/v1/grades/{level}/completion-status`	Completion status

Grade Assessments

Method	Endpoint	Description
GET	`/api/v1/grades/{level}/assessments`	List assessments per grade
POST	`/api/v1/grades/{level}/assessments/{id}/submit`	Submit assessment

Method	Endpoint	Description
GET	`/api/v1/grades/{level}/assessments/{id}/results`	Get results
GET	`/api/v1/grades/{level}/placement-test`	Placement test

Completion & Advancement

Method	Endpoint	Description
POST	`/api/v1/grades/{level}/complete`	Complete grade
GET	`/api/v1/grades/{level}/certificate`	Get certificate
POST	`/api/v1/grades/{level}/promotion`	Promote to next grade
GET	`/api/v1/grades/{level}/prerequisites`	Get prerequisites

Parent/Guardian Access (KG–Grade 8)

Method	Endpoint	Description
GET	`/api/v1/grades/students/{student_id}/progress`	View student progress
GET	`/api/v1/grades/students/{student_id}/report-card`	Get report card
POST	`/api/v1/grades/students/{student_id}/guardian-link`	Create guardian link
DELETE	`/api/v1/grades/students/{student_id}/guardian-link`	Delete guardian link

Admin - Grade Management

Method	Endpoint	Description
POST	`/api/v1/admin/grades/{level}/curriculum`	Create/edit curriculum
PUT	`/api/v1/admin/grades/{level}/lessons/{id}`	Update lesson
DELETE	`/api/v1/admin/grades/{level}/lessons/{id}`	Delete lesson
GET	`/api/v1/admin/grades/{level}/analytics`	Grade analytics
GET	`/api/v1/admin/grades/completion-rates`	Completion rates
GET	`/api/v1/admin/grades/students-by-grade`	Students by grade

Phase 5 — E-commerce, Admin & Scaling

Description
Cart and checkout; payment gateways (SkipCash, Sadad); orders and webhooks. Coupons, bundles, sponsorships, gifts; refunds. Support tickets and notifications; email templates. Admin: users, course approval, trainer applications, **dictionary admin**, **academic grade admin** (curriculum, lessons, analytics), SEO, popups, analytics. **Node app scaling:** stateless design, connection pooling, health/readiness for Kubernetes; horizontal scaling; optional Redis for cache/sessions (per SOW).

Timeline
Weeks 11–12 (Month 3).

Technical
Cart in DB or Redis; SkipCash/Sadad SDKs and webhook signature verification; 40/60 revenue-split calculation; SendGrid/SES for email templates; Bull/BullMQ for queues (optional); Redis for sessions/cache; health endpoints for K8s; no in-memory session state (stateless).

DB models
ShoppingCarts, CartItems, Orders, OrderItems, Invoices, Transactions; Coupons, CouponUsage, Bundles, BundleCourses, Sponsorships, Gifts; Refunds; SupportTickets, TicketMessages; EmailTemplates, Notifications; analytics/reporting tables per SOW.

Sub-phases

- **5.1** — Cart and checkout APIs; payment integration (SkipCash, Sadad); orders, verify webhook.
- **5.2** — Coupons, bundles, sponsorships, gift purchase/redeem; revenue split (40/60); refunds.
- **5.3** — Support tickets (create, reply, status, history); notifications; email template integration.
- **5.4** — Admin APIs: user management; course approval; trainer applications; **dictionary admin** (signs, categories, countries, Word of the Day, analytics); **academic grade admin** (grades overview, curriculum editor, lessons, assessments, analytics, completion rates, students-by-grade); SEO; popups; platform analytics.
- **5.5** — Node app scaling: stateless API, connection pooling, health/readiness for K8s; horizontal scaling; optional Redis (sessions/cache per SOW).
- **5.6** — **Testing:** Unit tests for cart, checkout, coupons, admin services; integration tests for checkout flow, payment webhook, refunds, support tickets.

API Endpoints

E-Commerce - Cart & Checkout

Method	Endpoint	Description
POST	`/api/v1/cart/add`	Add to cart
GET	`/api/v1/cart`	Get cart

Method	Endpoint	Description
PUT	`/api/v1/cart/update`	Update cart item
DELETE	`/api/v1/cart/remove`	Remove from cart
DELETE	`/api/v1/cart/clear`	Clear cart
POST	`/api/v1/cart/apply-coupon`	Apply coupon
DELETE	`/api/v1/cart/remove-coupon`	Remove coupon

Payments & Orders

Method	Endpoint	Description
POST	`/api/v1/checkout`	Checkout
POST	`/api/v1/payments/process`	Process payment
POST	`/api/v1/payments/verify`	Verify webhook (SkipCash/Sadad)
GET	`/api/v1/orders/{id}`	Get order
GET	`/api/v1/orders/history`	Order history
GET	`/api/v1/payments/gateways`	Available payment gateways

Coupons & Promotions

Method	Endpoint	Description
POST	`/api/v1/coupons`	Create coupon (Admin)
GET	`/api/v1/coupons`	List coupons (Admin)
GET	`/api/v1/coupons/{code}`	Get coupon details
PUT	`/api/v1/coupons/{code}`	Update coupon (Admin)
DELETE	`/api/v1/coupons/{code}`	Delete coupon (Admin)
POST	`/api/v1/coupons/{code}/validate`	Validate coupon
GET	`/api/v1/coupons/{code}/usage`	Coupon usage stats
POST	`/api/v1/coupons/bulk-create`	Bulk create (Admin)

Bundle Offers

Method	Endpoint	Description
POST	`/api/v1/bundles`	Create bundle (Admin)
GET	`/api/v1/bundles`	List bundles
GET	`/api/v1/bundles/{id}`	Get bundle details
PUT	`/api/v1/bundles/{id}`	Update bundle (Admin)
DELETE	`/api/v1/bundles/{id}`	Delete bundle (Admin)
POST	`/api/v1/bundles/{id}/purchase`	Purchase bundle
GET	`/api/v1/bundles/active`	Active bundles
GET	`/api/v1/bundles/{id}/courses`	Bundle courses

Gift System

Method	Endpoint	Description
POST	`/api/v1/gifts/purchase`	Purchase gift
GET	`/api/v1/gifts/sent`	Sent gifts
GET	`/api/v1/gifts/received`	Received gifts
POST	`/api/v1/gifts/redeem`	Redeem gift
GET	`/api/v1/gifts/{code}/status`	Gift status
PUT	`/api/v1/gifts/{code}/resend`	Resend gift
DELETE	`/api/v1/gifts/{code}/cancel`	Cancel gift

Refunds

Method	Endpoint	Description
POST	`/api/v1/refunds/request`	Request refund
GET	`/api/v1/refunds/my-requests`	My refund requests
GET	`/api/v1/refunds/{id}`	Get refund details
PUT	`/api/v1/refunds/{id}/approve`	Approve refund (Admin)
PUT	`/api/v1/refunds/{id}/reject`	Reject refund (Admin)
POST	`/api/v1/refunds/{id}/process`	Process refund (Admin)
GET	`/api/v1/refunds/eligible-orders`	Check eligibility

Trainer Earnings

Method	Endpoint	Description
GET	`/api/v1/earnings/dashboard`	Earnings dashboard
GET	`/api/v1/earnings/transactions`	Transaction history
GET	`/api/v1/earnings/payouts`	Payout history
POST	`/api/v1/earnings/request-payout`	Request payout
PUT	`/api/v1/earnings/payment-method`	Update payment method
GET	`/api/v1/earnings/analytics`	Earnings analytics
GET	`/api/v1/earnings/revenue-split`	Revenue split (40/60)
GET	`/api/v1/earnings/pending`	Pending earnings
GET	`/api/v1/earnings/paid`	Paid earnings

Support Tickets

Method	Endpoint	Description
POST	`/api/v1/tickets`	Create ticket
GET	`/api/v1/tickets/my-tickets`	My tickets
GET	`/api/v1/tickets/{id}`	Get ticket details
POST	`/api/v1/tickets/{id}/reply`	Reply to ticket

Method	Endpoint	Description
PUT	`/api/v1/tickets/{id}/status`	Update status
GET	`/api/v1/tickets/{id}/history`	Ticket history
POST	`/api/v1/tickets/{id}/close`	Close ticket
POST	`/api/v1/tickets/{id}/reopen`	Reopen ticket
POST	`/api/v1/tickets/{id}/attachments`	Add attachment

Admin - User Management

Method	Endpoint	Description
GET	`/api/v1/admin/users`	List all users
GET	`/api/v1/admin/users/{id}`	Get user details
PUT	`/api/v1/admin/users/{id}/suspend`	Suspend user
PUT	`/api/v1/admin/users/{id}/activate`	Activate user
DELETE	`/api/v1/admin/users/{id}`	Delete user
GET	`/api/v1/admin/users/{id}/activity`	User activity
GET	`/api/v1/admin/users/statistics`	User statistics
POST	`/api/v1/admin/users/bulk-action`	Bulk actions

Admin - Course Approval

Method	Endpoint	Description
GET	`/api/v1/admin/courses/pending`	Pending courses
GET	`/api/v1/admin/courses/under-review`	Under review
PUT	`/api/v1/admin/courses/{id}/approve`	Approve course
PUT	`/api/v1/admin/courses/{id}/reject`	Reject course
POST	`/api/v1/admin/courses/{id}/request-revision`	Request revision
GET	`/api/v1/admin/courses/{id}/review-history`	Review history
POST	`/api/v1/admin/courses/bulk-approve`	Bulk approve

Admin - Dictionary Management

Method	Endpoint	Description
POST	`/api/v1/admin/dictionary/signs`	Create sign
PUT	`/api/v1/admin/dictionary/signs/{id}`	Update sign

Method	Endpoint	Description
DELETE	`/api/v1/admin/dictionary/signs/{id}`	Delete sign
POST	`/api/v1/admin/dictionary/signs/{id}/videos`	Upload video
DELETE	`/api/v1/admin/dictionary/signs/{id}/videos/{videoId}`	Delete video
POST	`/api/v1/admin/dictionary/signs/{id}/assign-country`	Assign country
POST	`/api/v1/admin/dictionary/signs/bulk-import`	Bulk import
POST	`/api/v1/admin/dictionary/categories`	Create category
PUT	`/api/v1/admin/dictionary/categories/{id}`	Update category
DELETE	`/api/v1/admin/dictionary/categories/{id}`	Delete category
GET	`/api/v1/admin/dictionary/categories/tree`	Category tree
POST	`/api/v1/admin/dictionary/countries`	Add country
PUT	`/api/v1/admin/dictionary/countries/{code}`	Update country
DELETE	`/api/v1/admin/dictionary/countries/{code}`	Delete country
POST	`/api/v1/admin/dictionary/word-of-the-day/schedule`	Schedule word of day
PUT	`/api/v1/admin/dictionary/word-of-the-day/{date}`	Update scheduled
DELETE	`/api/v1/admin/dictionary/word-of-the-day/{date}`	Delete scheduled
GET	`/api/v1/admin/dictionary/word-of-the-day/upcoming`	Upcoming words
GET	`/api/v1/admin/dictionary/analytics/searches`	Search analytics
GET	`/api/v1/admin/dictionary/analytics/popular-signs`	Popular signs

Method	Endpoint	Description
GET	`/api/v1/admin/dictionary/analytics/by-country`	Analytics by country

Admin - Analytics & Reports

Method	Endpoint	Description
GET	`/api/v1/admin/analytics/overview`	Platform overview
GET	`/api/v1/admin/analytics/users`	User analytics
GET	`/api/v1/admin/analytics/courses`	Course analytics
GET	`/api/v1/admin/analytics/revenue`	Revenue analytics
GET	`/api/v1/admin/analytics/refunds`	Refund analytics
GET	`/api/v1/admin/analytics/enrollments`	Enrollment analytics
GET	`/api/v1/admin/analytics/trainers`	Trainer analytics
GET	`/api/v1/admin/reports/export`	Export report
POST	`/api/v1/admin/reports/custom`	Custom report
GET	`/api/v1/admin/reports/scheduled`	Scheduled reports

SEO Management

Method	Endpoint	Description
GET	`/api/v1/seo/pages`	List pages
GET	`/api/v1/seo/pages/{page}`	Get page SEO
PUT	`/api/v1/seo/pages/{page}/meta`	Update meta tags
GET	`/api/v1/seo/sitemap`	Get sitemap
PUT	`/api/v1/seo/sitemap/regenerate`	Regenerate sitemap
GET	`/api/v1/seo/robots`	Get robots.txt
PUT	`/api/v1/seo/robots`	Update robots.txt
GET	`/api/v1/seo/redirects`	List redirects
POST	`/api/v1/seo/redirects`	Create redirect
DELETE	`/api/v1/seo/redirects/{id}`	Delete redirect

Email Templates

Method	Endpoint	Description
GET	`/api/v1/email-templates`	List templates
GET	`/api/v1/email-templates/{id}`	Get template
PUT	`/api/v1/email-templates/{id}`	Update template
POST	`/api/v1/email-templates/{id}/test`	Test template
GET	`/api/v1/email-templates/{id}/logs`	Template logs
POST	`/api/v1/email-templates/{id}/preview`	Preview template

Method	Endpoint	Description
GET	`/api/v1/email-templates/categories`	Template categories

Popups & Banners

Method	Endpoint	Description
POST	`/api/v1/popups`	Create popup
GET	`/api/v1/popups`	List popups
GET	`/api/v1/popups/active`	Active popups
GET	`/api/v1/popups/{id}`	Get popup
PUT	`/api/v1/popups/{id}`	Update popup
DELETE	`/api/v1/popups/{id}`	Delete popup
GET	`/api/v1/popups/{id}/analytics`	Popup analytics
PUT	`/api/v1/popups/{id}/toggle-active`	Toggle active

Notifications

Method	Endpoint	Description
GET	`/api/v1/notifications`	Get notifications
GET	`/api/v1/notifications/{id}`	Get notification
PUT	`/api/v1/notifications/{id}/read`	Mark as read
PUT	`/api/v1/notifications/read-all`	Mark all as read
DELETE	`/api/v1/notifications/{id}`	Delete notification
GET	`/api/v1/notifications/unread-count`	Unread count
PUT	`/api/v1/notifications/preferences`	Update preferences
POST	`/api/v1/notifications/test`	Test notification (Admin)

File Upload

Method	Endpoint	Description
POST	`/api/v1/upload/image`	Upload image
POST	`/api/v1/upload/video`	Upload video
POST	`/api/v1/upload/document`	Upload document
POST	`/api/v1/upload/gif`	Upload GIF
POST	`/api/v1/upload/avatar`	Upload avatar
DELETE	`/api/v1/upload/{fileId}`	Delete file
GET	`/api/v1/upload/{fileId}/url`	Get file URL
POST	`/api/v1/upload/validate`	Validate file

Phase 6 — Testing & Documentation

Description
Automated tests and API documentation to meet SOW targets: e.g. 80%+ unit test coverage, integration tests for critical flows (auth, enrollment, payment, trainer application, dictionary, academic grades). Swagger/OpenAPI 3.0 and Postman collection for all modules.

Timeline
Weeks 13–14 (Month 4).

Technical
Jest (or Mocha); Supertest for HTTP; Swagger/OpenAPI 3.0 (swagger-jsdoc + swagger-ui-express); Postman collection export; coverage (e.g. 80%+ per SOW).

DB models
None (existing schema used for test data).

Sub-phases

- **6.1** — Unit tests (services, utils); integration tests (auth, enrollment, payment, trainer application, dictionary, academic grades).
- **6.2** — Swagger/OpenAPI 3.0 for all modules; Postman collection; error codes and auth flow documentation.

Phase 7 — Deployment

Description
Deploy backend to DigitalOcean: managed database, Spaces (if used), and Kubernetes (or App Platform). CI/CD, environment and secrets, SSL, monitoring, and runbook.

Timeline
Weeks 15–16 (Month 4).

Technical
Docker multi-stage build; GitHub Actions (or GitLab CI) for test + build; DigitalOcean: Managed PostgreSQL, Spaces (S3-compatible), Kubernetes cluster or App Platform; kubectl/Helm or DO App Spec; env/secrets (DO or K8s secrets); SSL (Let's Encrypt or DO); logging/monitoring (e.g. DO Monitoring, or external).

DB models
None (deployment and infrastructure only).

Sub-phases

- **7.1** — Dockerfile for Node API; CI pipeline (e.g. GitHub Actions) for build and test.
- **7.2** — DigitalOcean: managed DB (PostgreSQL), Spaces, secrets/env.

- **7.3** — Deploy to Kubernetes (or DO App Platform): deployments, services, ingress; env-specific config; health checks.
- **7.4** — SSL, domain, monitoring/logging; runbook and handoff notes.
- **7.5** — **Testing:** Smoke tests in staging; health-check and readiness verification; basic load check.

One-page summary

Month	Phases	Content
Month 1	1	Foundation, auth, DB (Users, auth tables).
Month 2	2, 3	Core (courses, enrollments, trainer). **Dictionary** (separate phase): search, browse, countries, Word of the Day, favorites, sign detail, GIF tooltips.
Month 3	4, 5	**Academic grade system** (separate). E-commerce, admin, dictionary/admin grade admin, Node scaling.
Month 4	6, 7	Testing and documentation. Deployment (DigitalOcean, K8s).

SOW alignment (backend only)

- **API architecture:** REST, JWT, `/api/v1/`, JSON, rate limiting, logging (Phase 1).
- **Modules:** Phase 1 (auth, users); Phase 2 (courses, enrollments, trainer apps); **Phase 3 (dictionary — separate)**; **Phase 4 (academic grade system — separate)**; Phase 5 (e-commerce, payments, coupons, bundles, sponsorships, gifts, refunds, tickets, admin including dictionary and grade admin, SEO, email templates, popups, notifications, file upload).
- **Future-phase modules (27–29):** EEG, AI, Healthcare — architecture only; not in this 4-month build.
- **Database:** 65+ tables per SOW across Phases 1–5; migrations version-controlled.
- **Performance:** Scaling in Phase 5.5; load testing and tuning in Phases 6 and 7.
- **Deployment:** Docker, CI/CD, DigitalOcean (DB, Spaces, K8s or App Platform), SSL, monitoring (Phase 7).

Document version: 1.2 | Backend API only | 4-month plan